Bank churn at a glance

This dataset covers **10,000 bank customers** and tracks whether each customer stayed or left the bank. The clearest headline is that churn is meaningful but concentrated: **20.4% of customers exited**, while **79.6% stayed**. Most of the variation comes from a few human factors such as **age, engagement, geography, gender, and product mix** rather than broad financial measures alone.

Exited	count
0	7963
1	2037

What the customer base looks like

The file includes customer details such as geography, gender, age, tenure, balance, number of products, card ownership, activity status, salary, and churn outcome. On average, customers are about **39 years old**, have a **credit score of 651**, and hold a **balance of 76.5k**.

CustomerId	Surname	CreditScore	Geography	Gender
15634602	Hargrave	619	France	Female
15647311	Hill	608	Spain	Female
15619304	Onio	502	France	Female
15701354	Boni	699	France	Female
15737888	Mitchell	850	Spain	Female

mean	std	min	25%	50%
15690940.5694	71936.18612274883	15565701.0	15628528.25	15690738.
650.5288	96.65329873613061	350.0	584.0	652.0
38.9218	10.487806451704591	18.0	32.0	37.0
5.0128	2.892174377049708	0.0	3.0	5.0

mean	std	min	25%	50%
76485.889288	62397.40520238623	0.0	0.0	97198.540
1.5302	0.5816543579989936	1.0	1.0	1.0
0.7055	0.4558404644751333	0.0	0.0	1.0
0.5151	0.49979692845891893	0.0	0.0	1.0
100090.239881	57510.49281769822	11.58	51002.11	100193.91
0.2037	0.4027685839948609	0.0	0.0	0.0

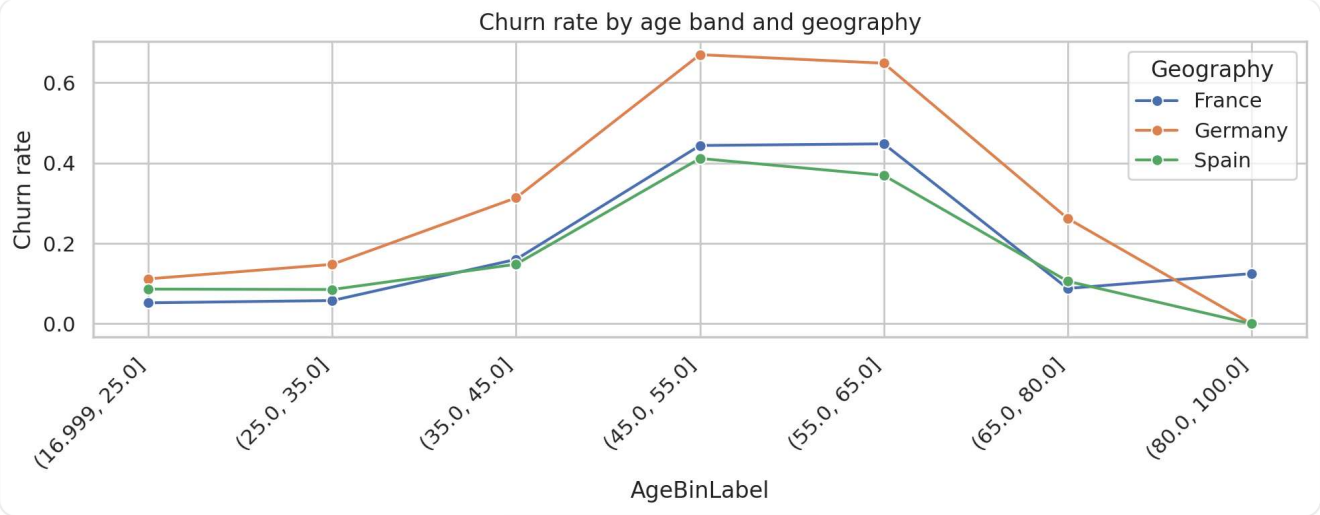
Age stands out as the strongest signal

Churn rises notably with age. Among all numeric fields, **age has the strongest positive relationship with churn (0.285)**, stronger than balance and far ahead of salary or credit score. The age distribution chart also shows churners skewing older than customers who stayed.

index	Exited
Exited	1.0
Age	0.285
Balance	0.119
EstimatedSalary	0.012
CustomerId	-0.006
HasCrCard	-0.007
Tenure	-0.014
CreditScore	-0.027
NumOfProducts	-0.048
IsActiveMember	-0.156

The pattern becomes even clearer when age is combined with location: churn climbs across older age bands in each geography, but not at the same pace everywhere.

Download



Geography makes a big difference

Where a customer lives matters. **Germany has the highest churn rate at 32.4%**, roughly double the level seen in **France (16.2%)** and well above **Spain (16.7%)**. This is one of the clearest splits in the dataset.

Geography	mean	count
Germany	0.32443204463929853	2509
Spain	0.1667339523617279	2477
France	0.16154766653370561	5014

Engagement is one of the biggest protective factors

Activity status sharply separates customers who stay from those who leave. **Inactive customers churn at 26.9%**, compared with **14.3% for active customers**. This is one of the strongest practical insights in the analysis and is also reflected in the correlation table, where activity has a notable negative relationship with churn.

IsActiveMember	mean	count
0	0.26850897092183956	4849
1	0.14269073966220153	5151

Product holding tells an especially sharp story

Customers with **two products** are the most stable group, with churn at just **7.6%**. Churn jumps to **27.7%** for customers with **one product**, becomes extremely high for **three products (82.7%)**, and reaches **100% for four products** in this sample. That makes product mix one of the most striking markers of risk.

NumOfProducts	mean	count
4	1.0	60
3	0.8270676691729323	266
1	0.27714398111723054	5084
2	0.07581699346405228	4590

Gender differences are visible, while cards and tenure matter less

Women churn more often than men in this dataset: **25.1% vs 16.5%**. By contrast, **having a credit card changes very little**, and **tenure differences are fairly small**, suggesting these are secondary factors compared with age, activity, geography, and number of products.

Gender	mean	count
Female	0.2507153863086066	4543
Male	0.16455928165658787	5457

HasCrCard	mean	count
0	0.20814940577249574	2945
1	0.20184266477675408	7055

Tenure	mean	count
0	0.23002421307506055	413
1	0.22415458937198068	1035
9	0.21646341463414634	984

Tenure	mean	count
3	0.21110009910802774	1009
5	0.20652173913043478	1012
10	0.20612244897959184	490
4	0.20525783619817997	989
6	0.20268872802481902	967
8	0.19219512195121952	1025
2	0.191793893129771	1048
7	0.17217898832684825	1028

Financial variables are less decisive on their own

Balance does have some relationship with churn, but it is modest compared with age and engagement. Salary barely moves the outcome, and credit score has only a weak negative relationship. The balance distribution chart supports this: there is some separation, but not enough to explain churn by itself.

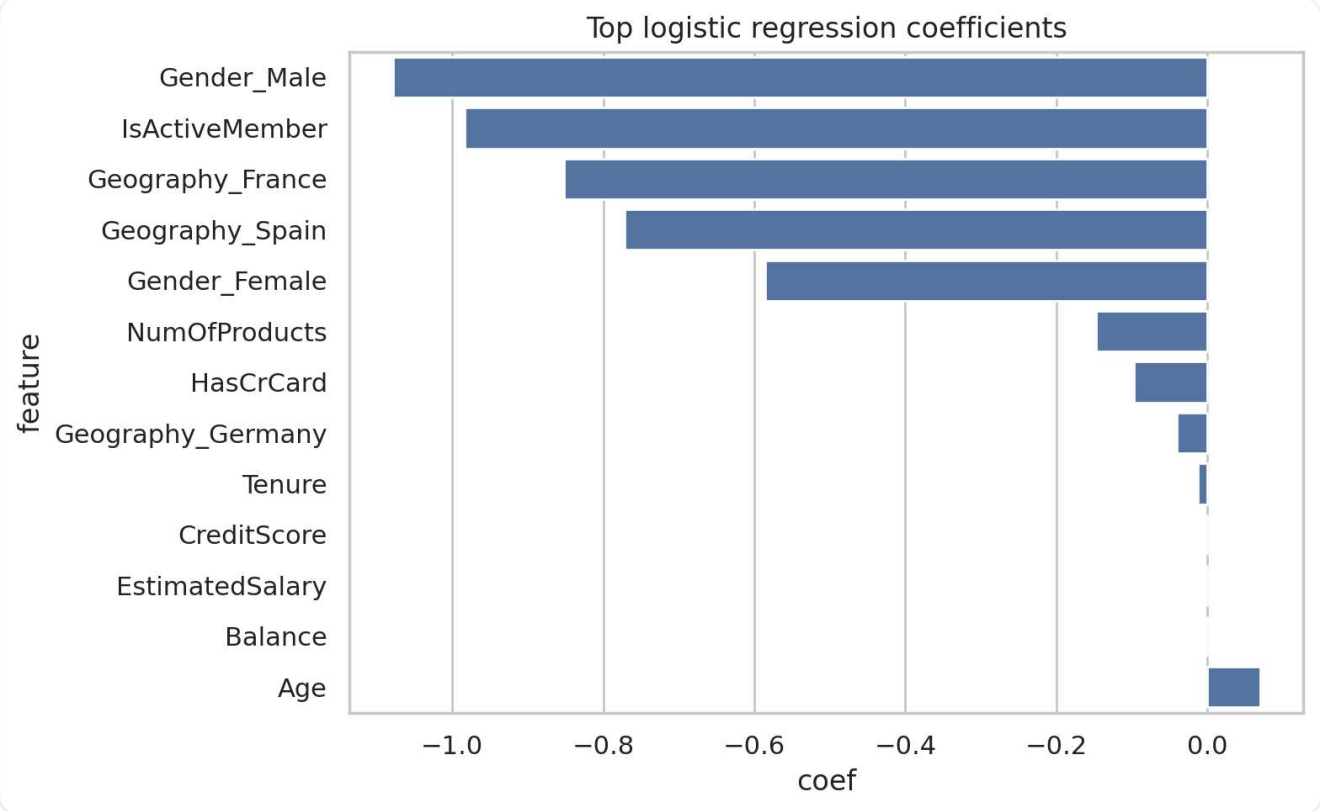
A simple model confirms the same story

A basic churn model performed reasonably well, with **AUC of 0.785**, which means it can separate higher-risk and lower-risk customers fairly well. The strongest directional signals in the model line up with the descriptive analysis: **activity status, gender, geography, age, and number of products** stand out most.

feature	coef	abs_coef
Gender_Male	-1.078312372579004	1.078312372579004
IsActiveMember	-0.9828343104347741	0.9828343104347741
Geography_France	-0.851361289377422	0.851361289377422
Geography_Spain	-0.7713582091688811	0.7713582091688811
Gender_Female	-0.585345522209047	0.585345522209047
NumOfProducts	-0.14728262219541896	0.14728262219541896

feature	coef	abs_coef
HasCrCard	-0.09768672111730642	0.09768672111730642
Age	0.0691521374006685	0.0691521374006685
Geography_Germany	-0.04093871563752746	0.04093871563752746
Tenure	-0.013101660089252492	0.013101660089252492
CreditScore	-0.0009073759243017545	0.0009073759243017545
Balance	2.5043563165980676e-06	2.5043563165980676e-06
EstimatedSalary	1.0522287134666037e-06	1.0522287134666037e-06

Download

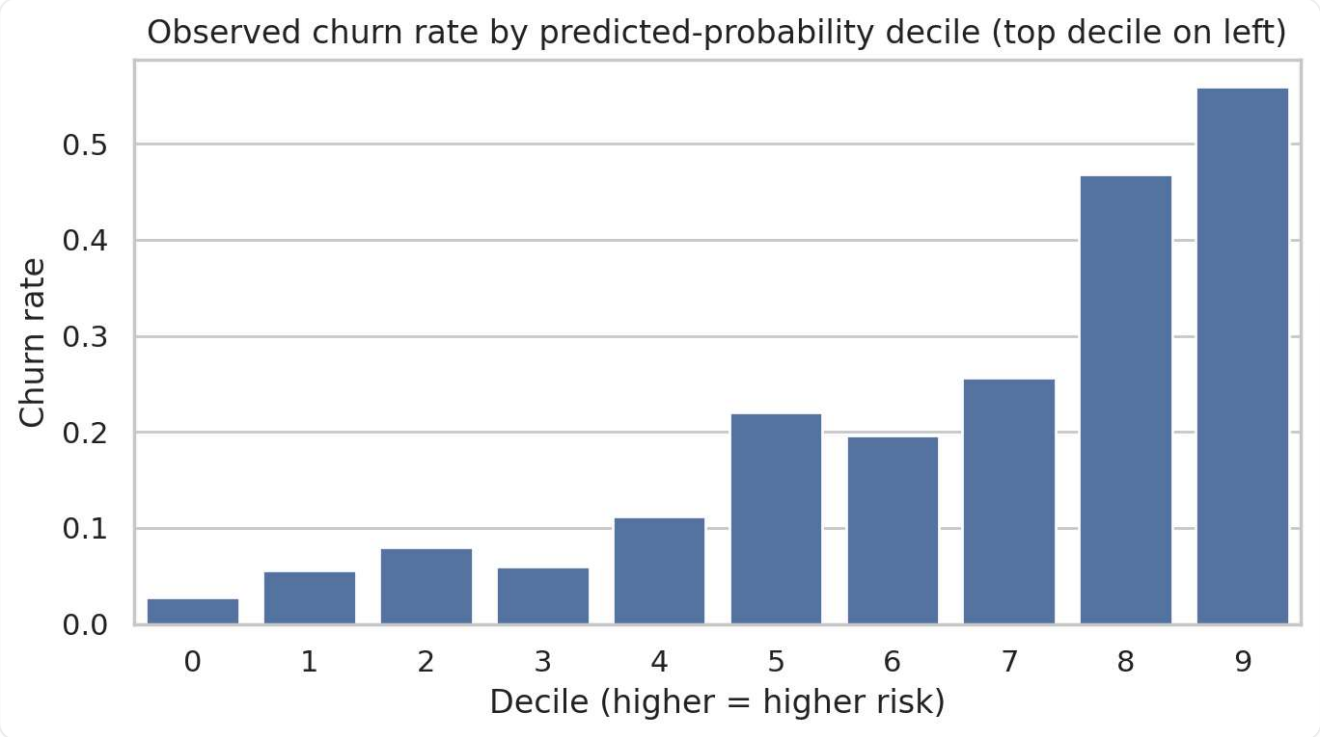


The risk ranking is also meaningful in practice: the **highest-risk decile has a churn rate of 56%**, while the **lowest-risk decile is just 2.8%**. That shows churn is not spread evenly across the base; it is concentrated in identifiable groups.

decile	churn_rate	n	avg_p
9	0.56	250	0.5642710016775786
8	0.468	250	0.3709692048999717

decile	churn_rate	n	avg_p
7	0.256	250	0.27354443311058085
6	0.196	250	0.21614256860890996
5	0.22	250	0.17440426666126632
4	0.112	250	0.1413330029344043
3	0.06	250	0.11385970755157103
2	0.08	250	0.08768905511621075
1	0.056	250	0.06379499126940788
0	0.028	250	0.03777363302726679

Download



Summary

The churn story is consistent across the analysis: **older customers, inactive customers, customers in Germany, women, and customers with unusual product counts are much more likely to leave**. Meanwhile, **credit cards, salary, and tenure add relatively little on their own**. In short, churn in this bank is driven more by **who the customer is and how engaged they are** than by simple financial profile alone.